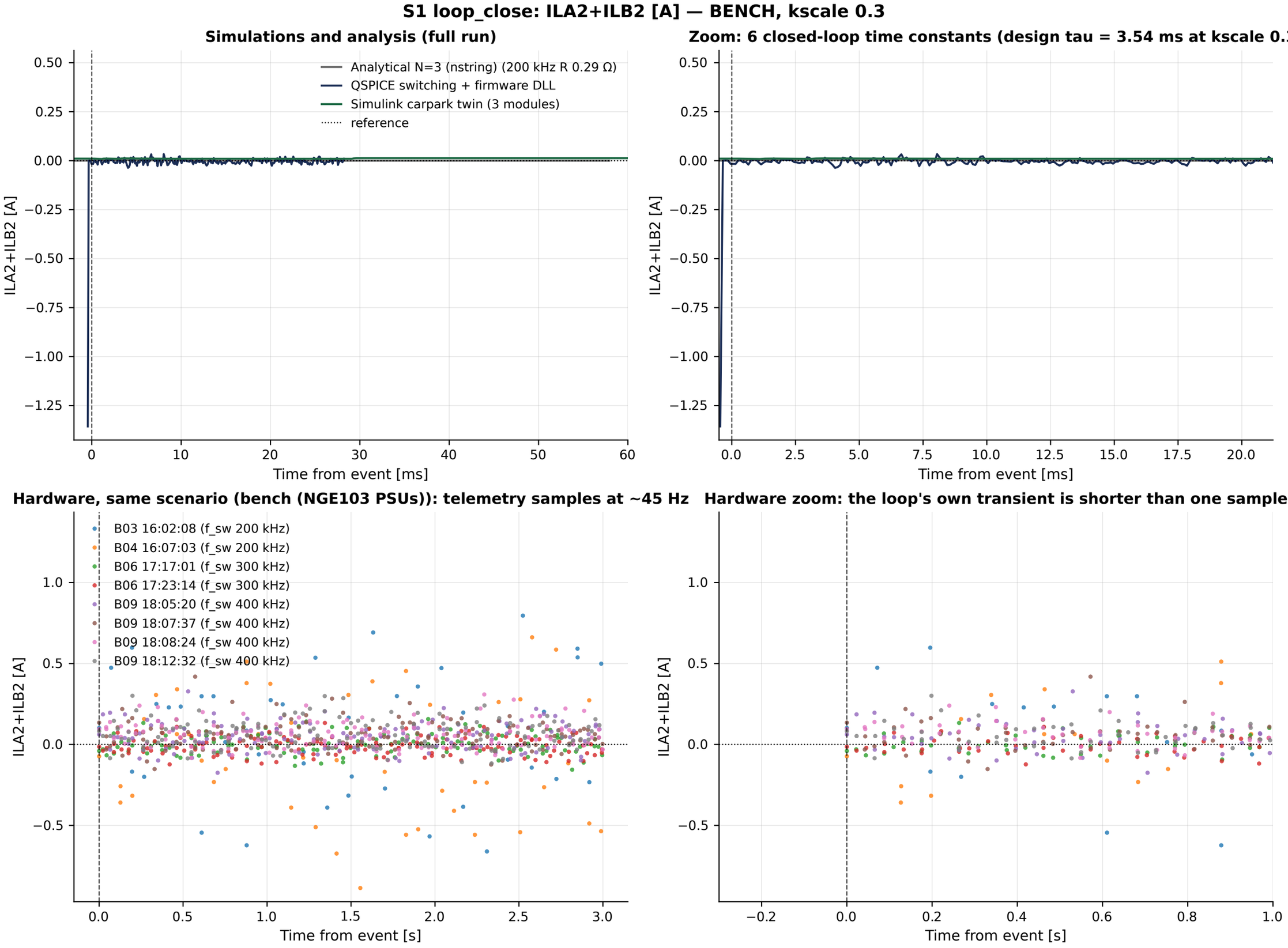

S1 loop_close: ILA2+ILB2 [A] — BENCH, kscale 0.3
Simulations and analysis (full run)
Zoom: 6 closed-loop time constants (design tau = 3.54 ms at kscale 0.
Analytical N=3 (nstring) (200 kHz R 0.29 Ω)
QSPICE switching + firmware DLL
Simulink carpark twin (3 modules)
reference
ILA2+ILB2 [A]
Time from event [ms]
Hardware, same scenario (bench (NGE103 PSUs)): telemetry samples at ~45 Hz
Hardware zoom: the loop's own transient is shorter than one sample
B03 16:02:08 (f_sw 200 kHz)
B04 16:07:03 (f_sw 200 kHz)
B06 17:17:01 (f_sw 300 kHz)
B06 17:23:14 (f_sw 300 kHz)
B09 18:05:20 (f_sw 400 kHz)
B09 18:07:37 (f_sw 400 kHz)
B09 18:08:24 (f_sw 400 kHz)
B09 18:12:32 (f_sw 400 kHz)
Time from event [s]